

PRODUCT REQUIREMENTS DOCUMENT

Desktop Accounting & POS System for Bhutanese Businesses

Version: 1.0 | **Status:** Production-ready | **Last Updated:** July 2026

Jinda Server: Licensing · Distribution · Admin | **Jinda Desktop:** Offline POS & Accounting

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Product Requirements Document (PRD)

Jinda POS Platform

Document Version: 1.0

Last Updated: July 2026

Product Name: Jinda POS (Desktop Accounting & POS System for Bhutan)

Platform: Jinda Server (licensing, distribution, admin) · Jinda Desktop App (offline POS & accounting)

Status: Production-ready (v1.0.0)

1. Product Overview

1.1 Product Vision

"Give every shop in Bhutan a powerful, offline, secure accounting & POS system that just works - with GST built-in, Bhutanese payment methods, and no monthly subscription stress."

1.2 Key Differentiators

- **Offline-first:** The desktop app works 100% offline. Sales, inventory, printing, reports, and accounting never stop because of internet problems.
- **Bhutan-specific:** Bhutanese Ngultrum (Nu.), 5% GST, domestic GST rate, Bhutanese payment methods (mBOB, BNB Pay, TPay, DrukPNB, BDBL, Druk Bank), dzongkhag/gewog address fields, Bhutanese-themed invoices.
- **Complete accounting:** True double-entry bookkeeping, chart of accounts, trial balance, P&L, balance sheet - not just a sales tool.
- **Secure by default:** Database encrypted at rest, tamper-evident audit logs, device-bound licenses, OS keychain token storage, strict CSP.
- **Fair licensing:** 7-day free trial (no credit card), one-year licenses, lifetime option, device transfer with OTP, 7-day offline grace.

2. Goals and Objectives

2.1 Business Goals

- 1 Provide Bhutanese SMEs with an affordable, reliable, offline POS & accounting system.
- 2 Generate revenue through license sales (Starter, Growth, Pro, Enterprise, Lifetime plans).
- 3 Build a sustainable update and support model with auto-updates.
- 4 Reduce software piracy via device-bound licenses with OTP-gated transfers.
- 5 Establish Jinda as the leading Bhutanese accounting & POS software brand.

2.2 Product Goals

- 1 **Usability:** A non-technical shop owner can install, activate, and make their first sale within 10 minutes.
- 2 **Reliability:** The app must not crash, lose data, or stop working due to internet outages.

- 3 Compliance:** GST is calculated and reported correctly per Bhutanese tax law (5% standard, effective January 2026).
- 4 Security:** Customer financial data is encrypted, audited, and protected from tampering.
- 5 Maintainability:** The server admin can create licenses, publish updates, and manage customers without developer help.

2.3 Measurable Objectives

Objective	Target	Measurement
First-sale time (onboarding to first sale)	≤ 10 minutes	User testing
App uptime (offline operation)	100% (no internet dependency for daily ops)	Crash reports
License activation success rate	≥ 99%	Server logs
Update adoption within 7 days	≥ 80% of eligible installs	<code>/api/admin/security/dashboard</code>
Critical bug count at release	0	Pre-release QA
GST report accuracy	100% match with manual calculation	Accounting audit
DB encryption at rest	100% of installs	Integrity verification on boot

3. Target Users and Personas

3.1 Primary Personas

Persona 1 - Tashi, the Shop Owner (Admin User)

- Owns a small grocery or retail shop in Thimphu.
- Not very technical; wants something that "just works".
- Needs: sales, inventory, GST, daily reports, low-stock alerts.
- Pain: current solutions need internet, don't do GST right, or are too expensive.

Persona 2 - Deki, the Cashier (Staff User)

- Works at Tashi's shop, operates the POS daily.
- Needs: fast item search, barcode scanning, quick billing, thermal printing.
- Does not need access to reports, settings, or voiding.

Persona 3 - Karma, the Accountant

- Handles GST filing and year-end accounts for multiple shops.
- Needs: trial balance, P&L, balance sheet, GST summary, Excel/PDF export.
- Wants: clean, accurate, printable reports.

Persona 4 - Pema, the Server Admin (Internal)

- Manages licenses, customers, updates, and security for the Jinda platform.
- Needs: admin dashboard, license creation, revocation, update publishing, security monitoring.

3.2 Secondary Personas

- **A growing business owner** who needs multi-user (2 - 5 staff) and tiered pricing.
- **A franchise owner** who needs branch management.
- **A consultant** who imports data via Excel for new clients.

3.3 Anti-Personas (Not the Target)

- Large enterprises needing cloud-based multi-branch real-time sync (Jinda is offline-first, single-machine).
- Users outside Bhutan (the product is Bhutan-specific in tax, currency, and payments).
- Users wanting a web-only SaaS (the app is a desktop app by design).

4. Market and Problem Statement

4.1 The Problem

Bhutanese small businesses struggle with accounting & POS software because:

- 1 **Most software needs constant internet** - but internet in Bhutan can be unreliable and expensive. Shops lose sales when the internet goes down.
- 2 **Foreign software doesn't fit Bhutan**: wrong currency (USD), wrong tax (no 5% GST), wrong payment methods (no mBOB/BNB/TPay), wrong address format (no dzongkhag/gewog).
- 3 **Existing solutions are either too simple** (just a billing app, no real accounting) **or too complex** (full ERP, needs a trained accountant).
- 4 **Piracy is common** - shop owners share one license across many computers, hurting software makers.
- 5 **Data security is poor** - most local tools store data in plaintext, so a stolen laptop means stolen financial data.
- 6 **Monthly subscriptions are a burden** for small shops with thin margins.

4.2 The Solution

Jinda solves all of these:

- 1 **Offline-first** - works without any internet for daily operations.
- 2 **Built for Bhutan** - Nu. currency, 5% GST, Bhutanese payment methods, dzongkhag/gewog addresses, Bhutanese-themed invoices.
- 3 **Complete but simple** - double-entry accounting under the hood, but the shop owner just clicks "Make Sale".
- 4 **Device-bound licenses with OTP transfers** - stops piracy while allowing legitimate device changes (max 2/month).
- 5 **Encrypted database at rest** - AES-256 + HMAC; stolen laptop = unreadable data.
- 6 **One-time license purchase** - Starter is a 1-year license; Lifetime option available; no monthly fees.

5. Product Scope

5.1 In Scope (v1.0)

Server (Licensing & Distribution Platform):

- Public marketing website (home, features, pricing, about, contact, download, docs).
- License lifecycle: creation, activation, device binding, transfer (OTP), revocation, extension.
- POS user accounts: registration, email verification, login, device verification, password reset, sessions with refresh-token rotation.
- Admin dashboard: licenses, customers, users, updates, messages, security dashboard, settings.
- Software update distribution (electron-updater feed + JSON API).
- Email sending (verification, OTP, reset, license transfer OTP, contact replies).
- AI chatbot (OpenRouter).
- Rate limiting (Redis + memory fallback).
- Audit logging.
- Security headers (CSP, HSTS, etc.).

Desktop App (POS & Accounting):

- Onboarding: EULA, trial, license activation, initial admin account creation.
- Authentication: local login (admin/staff), plan seat limits.
- Dashboard: real-time metrics, alerts, recent transactions.
- POS: cart, barcode, GST auto-calc, split payments, credit control, held carts, email invoice, A4 + thermal printing.
- Inventory: items, categories, units, stock, average cost, low-stock, barcode mapping.
- Contacts: customers & suppliers with auto-ledgers, credit limits, tiered pricing.
- Transactions: receive/pay/transfer money, void (admin), export.
- Double-entry accounting engine with chart of accounts.
- Purchase orders (with PO-driven selling price).
- Quotations (convert to sale).
- Refunds (with reversal entries).
- Expenses.
- Recurring transactions (hourly processing).
- GST: 5% standard, 0% domestic, monthly summary, returns filing, period close.
- Reports: trial balance, P&L, balance sheet, outstanding, stock, sales, payroll, purchase, customer insights, aged receivables/payables.
- HR: employees (PF/GIS/TDS/HC), payroll, branches.
- Exports: Excel (xlsx) and PDF (jsPDF) with Bhutanese theme.
- Imports: CSV/Excel for contacts and items.
- Audit log (tamper-evident).
- Backups: local (daily auto), restore, Google Drive cloud, MEGA cloud.
- Database encryption at rest.
- Auto-update (electron-updater, 4-hour check, SHA512 verify).
- Settings: company info, GST rates, logo/seal/signature, users, period close, EULA.

5.2 Out of Scope (v1.0)

See Section 20.

6. Success Metrics

6.1 Adoption Metrics

- Number of trial starts per month.
- Trial-to-paid conversion rate (target: $\geq 30\%$).
- Number of active licenses.
- License renewal rate (target: $\geq 70\%$).
- Number of downloads per month.

6.2 Engagement Metrics

- Average sales per day per active install.
- Number of GST returns filed via the app per month.
- Number of reports exported per month.
- Update adoption rate within 7 days (target: $\geq 80\%$).

6.3 Quality Metrics

- Crash-free sessions (target: $\geq 99.5\%$).
- License activation success rate (target: $\geq 99\%$).
- Support tickets per 100 users per month (target: ≤ 5).
- Critical bugs in production (target: 0).
- GST calculation accuracy (target: 100%).

6.4 Business Metrics

- Monthly recurring revenue (from new license sales).
- Customer acquisition cost.
- Lifetime value per customer.
- Refund rate (target: $\leq 5\%$).

7. Assumptions and Constraints

7.1 Assumptions

- Users have a Windows 10/11 computer (64-bit) at their shop.
- Users have a printer (A4 or thermal) for invoices.
- Users have an email address for license delivery and OTP.
- The Bhutan GST rate remains 5% (standard) and 0% (domestic) unless changed by law.
- Internet is available at least once every 7 days (the offline grace period).
- The server is hosted on Vercel with MongoDB Atlas and optional Redis.

7.2 Constraints

- **Offline-first:** The desktop app must function fully without internet (except for license checks and updates).
- **Single-machine:** Each license binds to one device; data is not synced between machines in v1.0.
- **Bhutan only:** Currency, tax, and payment methods are Bhutan-specific.
- **Electron:** The desktop app is built on Electron (not a native app or web app).

- **SQLite:** The local database is SQLite (not a server database) for offline operation.
- **Windows-first:** While macOS and Linux builds are supported, Windows is the primary target.
- **License transfer limit:** Maximum 2 device transfers per month per license.
- **Plan seat limits:** Starter=1, Growth/Pro=2, Enterprise/Lifetime=5 users.

7.3 Dependencies

- MongoDB Atlas (server database).
- Redis (optional; rate limiting with memory fallback).
- Vercel (server hosting).
- Cloudflare R2 or AWS S3 (optional; installer file storage).
- GitHub Releases (update file hosting).
- Gmail SMTP (email delivery).
- OpenRouter (AI chatbot; optional).
- Google Drive API & MEGA API (cloud backup).
- Node.js 20+, Python 3, VS Build Tools C++ (for building the desktop app).

8. Licensing and Plans

See SOP Sections 3.1 - 3.5 for the complete plan definitions, license key format, license states, license/trial rules, and pricing details.

>

Summary: The system supports 6 plan tiers (Free Trial 7-day, Starter, Growth, Pro, Enterprise, Lifetime) with seat limits of 1 - 5 users. Licenses follow a **DTS-XXXX-XXXX-XXXX** format with states: inactive, active, expired, suspended, revoked. Device transfer is gated by OTP (max 2/month). The app enforces a 7-day offline grace period after last successful verification.

9. Functional Requirements - Server

9.1 Public Website

FR-S01 - Landing Page

The system shall display a landing page at **/** with hero, features, pricing preview, and call-to-action to download or start trial.

FR-S02 - Feature Pages

The system shall display pages for **/features**, **/pricing**, **/about**, **/contact**, **/docs**, **/download**, **/security**, **/privacy**, **/terms**, **/refund**, **/license**, **/license-activate**.

FR-S03 - Download Page

The system shall provide download links for the Windows installer (NSIS) and portable version, sourced from GitHub Releases or R2.

FR-S04 - Contact Form

The system shall accept contact form submissions at **/api/contact**, store them in MongoDB (**ContactMessage**), and email the admin. All input shall be XSS-escaped.

FR-S05 - SEO

The system shall include sitemap, robots.txt, OpenGraph/Twitter images, JSON-LD, and per-page metadata. Admin and sensitive pages shall have **X-Robots-Tag: noindex**.

FR-S06 - AI Chatbot

The system shall provide an AI chatbot at **/api/chat** that streams responses via SSE from OpenRouter, with 8 free fallback models and a 30 req/min rate limit.

9.2 Authentication (POS Users)**FR-S07 - Registration**

The system shall allow POS user registration at **/api/auth/register** with email, username, and a strong password. Password is bcrypt-hashed (12 rounds). A verification email with a 24h-valid token is sent.

FR-S08 - Email Verification

The system shall verify emails via **/api/auth/verify-email** (POST with token, or GET via link). On success, **isVerified** is set, **accountStatus** updates, and a session is auto-created.

FR-S09 - Login

The system shall authenticate POS users at **/api/auth/login** with rate limiting (5/15min) and account lockout (5 failed attempts -> 15-min lock). On new device, a 6-digit OTP is emailed.

FR-S10 - Device Verification

The system shall verify device OTPs at **/api/auth/verify-device** (3 attempts -> 15-min lock). On success, the device is added to **approvedDevices** and a session is created. Resend rate-limited (1/30s).

FR-S11 - Password Reset

The system shall support forgot-password (**/api/auth/forgot-password**) and reset-password (**/api/auth/reset-password**) with anti-enumeration (does not reveal if an email exists). Reset revokes all sessions.

FR-S12 - Password Change

The system shall support authenticated password change at **/api/auth/change-password**, revoking other sessions.

FR-S13 - Session Management

The system shall issue short-lived access tokens (15-min JWT) and refresh tokens (7-day, SHA-256 hashed in DB). Refresh-token rotation shall detect reuse and revoke the entire token family.

FR-S14 - Captcha

The system shall provide math/text captcha at **/api/auth/captcha** for forms.

9.3 Licensing**FR-S15 - License Verification & Activation**

The system shall validate and activate licenses at **/api/license/verify**. First activation binds the device and returns **activationSecret**. Same-device re-handshake increments **activationCount**. Different-device triggers the transfer flow.

FR-S16 - Device Transfer

The system shall allow device transfers up to 2/month. Step 1: user provides password -> server verifies (bcrypt) and emails a 6-digit OTP. Step 2: user provides OTP -> server rebinds the device and rotates

`activationSecret`.

FR-S17 - License Activation (Light)

The system shall provide `/api/license/activate` for lighter activation (validates, rejects if on another device, binds device, increments count).

FR-S18 - License Revocation

The system shall allow admin to revoke a license at `/api/license/revoke` with a validated reason. Revocation stamps `revokedAt`, `revokedBy`, `revokedReason`, `revokedNotes` and logs a critical audit event.

9.4 Admin Dashboard

FR-S19 - Admin Login

The system shall provide admin login via NextAuth (credentials provider, JWT session, 24h maxAge) at `/admin/login`. Only `role==='admin'` users can access admin pages.

FR-S20 - Dashboard

The system shall display license counts, plan breakdowns, expiring-soon licenses, and recent licenses at `/admin/dashboard` via `/api/stats`.

FR-S21 - License CRUD

The system shall allow admins to create (`/api/admin/licenses` POST), list (GET, paginated + search), view, update (PATCH), delete, and extend licenses.

FR-S22 - Customer CRUD

The system shall allow admins to create, list, view, update, and delete customers (delete blocked if active licenses exist).

FR-S23 - User Management

The system shall allow admins to list POS users and perform actions: activate, suspend, terminate, disable, lock, unlock, reset_password, verify_email via `/api/admin/users`.

FR-S24 - Update Management

The system shall allow admins to create, list, edit, delete, and change status of software updates. Statuses: draft, published, blocked, rollback. Supports rollout % and forced flag.

FR-S25 - Message Management

The system shall allow admins to list contact messages, update status (Unread/Read/Replied), and reply by email via `/api/admin/messages/[id]/reply`.

FR-S26 - Security Dashboard

The system shall display security metrics at `/admin/security` via `/api/admin/security/dashboard`: alerts, login stats, active sessions, locked accounts, top failing IPs, suspicious users.

9.5 Update Distribution

FR-S27 - Latest Update API

The system shall provide `/api/updates/latest` returning the latest published update as JSON (respecting rollout %).

FR-S28 - electron-updater YAML

The system shall provide `/updates/latest.yml` in electron-updater format (version, files, sha512, size, path, releaseDate).

FR-S29 - Update Upload

The system shall allow admins to register an update from a GitHub Release URL at `/api/updates/upload` (files hosted on GitHub Releases).

9.6 Email

FR-S30 - Email Sending

The system shall send emails via Nodemailer (Gmail SMTP): verification emails, reset emails, license transfer OTPs, device verification OTPs, and contact reply emails.

FR-S31 - Email URL

Verification and reset links shall use `NEXTAUTH_URL` (not `VERCEL_URL`) so links are stable across deployments.

9.7 Health and Legal

FR-S32 - Health Check

The system shall provide `/api/health` returning `{ status: "ok", timestamp }` for the desktop app to check server reachability.

FR-S33 - Terms & Privacy

The system shall provide `/api/terms` and `/api/privacy` returning JSON for the desktop app's EULA window.

9.8 Rate Limiting & Audit

FR-S34 - Rate Limiting

The system shall rate-limit endpoints: default 10/min, license 5/min, API 60/min, auth 5/15min. Redis-backed with memory fallback. Headers: `Retry-After`, `X-RateLimit-*`.

FR-S35 - Audit Logging

The system shall log 29 audit event types (login success/failure, token reuse, license activated/transferred/revoked, device bound/mismatch, account locked, brute force, admin actions, etc.) with severity (info/warning/critical) to MongoDB.

10. Functional Requirements - Desktop App

10.1 Onboarding

FR-D01 - EULA

The app shall display the End-User License Agreement on first launch. Acceptance is saved locally. If the terms version changes, the user must re-accept.

FR-D02 - Welcome Screen

The app shall show a Welcome screen with two options: Start 7-day Free Trial, or Activate a License.

FR-D03 - Trial Start

The app shall start a 7-day trial bound to the device ID, writing `trial.json` and an encrypted `.trial-lock`. Trial cannot be reset by deleting files.

FR-D04 - License Activation

The app shall activate a license by sending the key + device ID to the server. It shall handle the multi-step transfer flow (CREDENTIALS_REQUIRED -> OTP_REQUIRED -> valid).

FR-D05 - Initial Admin Setup

The app shall require creation of an initial admin account (username + strong password) after activation/trial.

FR-D06 - Email Verification (POS SaaS)

If the user registered via the website, the app shall gate on email verification and device verification before granting access.

10.2 Authentication & Authorization

FR-D07 - Login

The app shall authenticate local users (admin and staff) with username + password (bcrypt-verified against the local SQLite `users` table).

FR-D08 - Role-Based Access

The app shall restrict admin-only pages (Settings, Reports, Payroll, Void, Backup, Users) to admin users. Staff users are redirected to the POS.

FR-D09 - Seat Limits

The app shall enforce plan seat limits. The admin seat is free. Only active non-admin staff count. Creating a user beyond the limit is blocked.

10.3 Dashboard

FR-D10 - Dashboard

The app shall display a dashboard (admin only) with today's sales, GST collected, transaction count, low-stock alerts, and recent transactions.

FR-D11 - Real-Time Metrics

The app shall update dashboard metrics in real time as sales occur.

FR-D12 - Notifications

The app shall display header alerts (low stock, trial ending, update available) and toast notifications (with an audio chime, mutable).

10.4 POS (Point of Sale)

FR-D13 - Cart

The app shall provide a cart with item search (by name, code, barcode), quantity, per-line GST, line totals, and cart total.

FR-D14 - Barcode Scanning

The app shall support barcode scanning to find items (via the `barcode_mappings` table).

FR-D15 - GST Auto-Calculation

The app shall auto-calculate GST per line at the item's GST rate (default 5%). If tax type is "domestic", the domestic GST rate (default 0%) is used.

FR-D16 - Discounts

The app shall support discounts on the cart or per line.

FR-D17 - Payment Modes

The app shall support: Cash, Bank, Credit, Card, UPI, mBOB, BNB, TPay, DrukPNB, BDBL, DKBank.

FR-D18 - Split Payments

The app shall support split payments (multiple payment methods for one sale).

FR-D19 - Credit Control

The app shall block credit sales when a customer exceeds their credit limit.

FR-D20 - Held Carts

The app shall allow holding (parking) a cart and resuming it later. Held carts persist in the `held_carts` table.

FR-D21 - Complete Sale

On sale completion, the app shall: reduce inventory, update customer ledger (if credit), create a double-entry transaction, write a GST entry, and write an audit log.

FR-D22 - Printing

The app shall print A4 invoices (Standard, Modern, Professional templates) and 80mm thermal receipts, using the native Windows print dialog.

FR-D23 - Email Invoice

The app shall allow emailing an invoice to the customer (via the system mail client).

10.5 Inventory

FR-D24 - Item Management

The app shall allow creating, updating, and deleting items with: code, name, category, unit, purchase price, selling price, opening stock, reorder level, GST applicable, GST rate.

FR-D25 - Stock Management

The app shall track stock movements (in, out, adjustment) and auto-calculate average cost.

FR-D26 - Low-Stock Alerts

The app shall flag items below their reorder level on the dashboard and inventory page.

FR-D27 - Categories & Units

The app shall allow creating custom item categories and units.

FR-D28 - Barcode Mapping

The app shall allow mapping barcodes to items for scanning at the POS.

10.6 Contacts

FR-D29 - Customers & Suppliers

The app shall manage customers (`type=customer`) and suppliers (`type=supplier`) with name, contact person, phone, email, address (street, gewog, dzongkhag), credit limit, credit days, opening balance, GST number.

FR-D30 - Auto-Ledger

The app shall auto-create a ledger account for each contact and track their current balance.

FR-D31 - Customer Statement

The app shall display a per-customer ledger (invoices, payments, balance).

FR-D32 - Tiered Pricing

The app shall support price lists assigned to customers (special prices per item per customer).

10.7 Transactions & Accounting

FR-D33 - Receive Money

The app shall record money received from customers (reduces receivable).

FR-D34 - Pay Money

The app shall record money paid to suppliers (reduces payable).

FR-D35 - Transfer

The app shall support transfers between accounts (e.g., Cash -> Bank).

FR-D36 - Void

The app shall allow admin to void a transaction (with a reason), creating a reversal entry. Voided transactions are excluded from reports and GST.

FR-D37 - Double-Entry Engine

The app shall maintain a double-entry accounting system where every transaction has balanced debits and credits in **transaction_lines**.

FR-D38 - Chart of Accounts

The app shall seed a standard Bhutanese chart of accounts (Assets, Liabilities, Equity, Income, COGS, Expenses) and allow adding custom accounts.

10.8 Procurement

FR-D39 - Purchase Orders

The app shall allow admins to create POs (supplier, items, quantities, purchase price, selling price), receive them (increases stock, updates average cost), and pay suppliers. POs support a **selling_price** column for PO-driven pricing.

FR-D40 - Quotations

The app shall allow creating quotations for customers and converting them to sales in one click.

FR-D41 - Refunds

The app shall allow admins to create refunds against original invoices, with reversal journal entries and GST adjustments. Refunded stock returns to inventory.

10.9 Finance

FR-D42 - Expenses

The app shall allow admins to track expenses (date, category, amount, payment mode, vendor, receipt path), each creating a transaction.

FR-D43 - Recurring Transactions

The app shall allow admins to schedule recurring transactions (daily, weekly, monthly, quarterly, yearly), processed hourly and on startup.

FR-D44 - Aged Reports

The app shall provide aged receivables and aged payables reports with buckets (0 - 30, 31 - 60, 61 - 90, 90+ days).

10.10 GST

FR-D45 - GST Calculation

The app shall calculate GST at 5% (standard) or 0% (domestic) per the tax type on each invoice/line.

FR-D46 - GST Entries

The app shall write `gst_entries` (type input/output, amount, rate, month, year, is_filed) for every taxable transaction.

FR-D47 - GST Summary

The app shall provide a monthly GST summary: taxable sales, taxable purchases, GST output, GST input, exempt, net payable.

FR-D48 - GST Return Filing

The app shall allow admins to file a GST return for a month/year, marking entries as filed.

FR-D49 - GST Rates Integrity-Sealed

The app shall integrity-seal the `gst_rate` and `gst_rate_domestic` settings to prevent direct DB tampering.

FR-D50 - Period Close

The app shall allow admins to lock a period (year + month), preventing transaction edits. Period locks are integrity-sealed.

10.11 Reports

FR-D51 - Trial Balance

The app shall generate a trial balance (all accounts with debit/credit balances; must balance).

FR-D52 - Profit & Loss

The app shall generate a P&L statement (income vs. expenses -> net profit/loss) for a period.

FR-D53 - Balance Sheet

The app shall generate a balance sheet (assets, liabilities, equity) as of a date.

FR-D54 - Outstanding Report

The app shall generate an outstanding receivables/payables report.

FR-D55 - Stock Report

The app shall generate a stock report (items, quantities, value at average cost, low/out of stock).

FR-D56 - Sales Report

The app shall generate a sales report with summary cards (revenue, GST, transaction count).

FR-D57 - Payroll Report

The app shall generate a payroll report (salary payments by period).

FR-D58 - Purchase Report

The app shall generate a purchase report by period.

FR-D59 - Customer Insights

The app shall generate per-customer insights (sales, balances, top customers).

10.12 HR

FR-D60 - Employees

The app shall allow admins to manage employees (employee no, name, position, department, phone, email, salary, join date, PF rate, GIS amount, TDS rate, HC rate).

FR-D61 - Payroll

The app shall allow admins to process payroll (calculate gross, PF, TDS, GIS, HC, net) and create payment transactions.

FR-D62 - Branches

The app shall allow admins to manage branches (name, code, address, phone, email).

10.13 Imports & Exports**FR-D63 - CSV/Excel Import**

The app shall allow admins to import contacts and items from CSV/Excel files (bounded to 10,000 rows).

FR-D64 - Excel Export

The app shall export to Excel (.xlsx) with Bhutanese-themed styling: transactions, contacts, items, expenses, employees, quotations, refunds, purchase orders, aged reports, trial balance, P&L, balance sheet, outstanding, stock, sales.

FR-D65 - PDF Export

The app shall export reports to PDF (jsPDF + autotable) with the Bhutanese theme.

10.14 Audit**FR-D66 - Audit Log**

The app shall maintain a tamper-evident audit log (**audit_logs**) of all create/update/delete/void/login actions, with old/new values, user, IP, timestamp. The chain is verified on every boot.

FR-D67 - Audit Log Viewer

The app shall provide an admin-only audit log viewer.

10.15 Backup & Restore**FR-D68 - Local Backup**

The app shall allow admins to create and restore local backups (streaming JSON, memory-bounded). The app relaunches after restore.

FR-D69 - Auto Backup

The app shall auto-back up daily at 23:00.

FR-D70 - Emergency Backup

The app shall maintain **backups/emergency_*.db** files and use them for crash recovery if the main DB fails to open.

FR-D71 - Cloud Backup (Google Drive)

The app shall allow admins to connect Google Drive via OAuth (loopback server, CSRF nonce) and back up to / restore from Drive.

FR-D72 - Cloud Backup (MEGA)

The app shall allow admins to connect MEGA and back up to / restore from MEGA.

10.16 Database Security**FR-D73 - Encryption at Rest**

The app shall encrypt the SQLite database to a **.secure** file (AES-256-CBC + HMAC) on close and decrypt on open. The plaintext **.db** is securely deleted when closed.

FR-D74 - Integrity Seals

The app shall integrity-seal (HMAC-SHA256) critical values: GST rates, user roles, period locks. Seals are verified on every boot; violations logged as **SECURITY_ALERT** audit events.

FR-D75 - Crash Recovery

The app shall recover from a crash by preferring a valid plaintext **.db**, else decrypting **.secure**, else restoring from the newest **emergency_*.db**.

10.17 License (Desktop Side)

FR-D76 - License Status

The app shall compute and display the license status: **trial**, **licensed**, **trial_expired**, **license_expired**, **license_device_mismatch**, **verification_required**, **none**.

FR-D77 - Periodic Verification

The app shall verify the license with the server 1 minute after boot, then every 24 hours. Within the 7-day offline grace, the app remains usable.

FR-D78 - Trial Banner

The app shall display a trial banner showing days remaining.

10.18 Auto-Update

FR-D79 - Update Check

The app shall check for updates 30 seconds after launch, then every 4 hours, via the server's update feed.

FR-D80 - Update Download

The app shall download updates automatically in the background and verify the SHA512 hash before marking complete.

FR-D81 - Update Banner

The app shall display an update banner when a download is complete.

FR-D82 - Update Install (Admin Only)

Only admin users shall install updates. Before installing, the app shall create a pre-update backup.

10.19 Settings

FR-D83 - Company Settings

The app shall allow admins to set: company name, trade license no, tax no, address, phone, email, default GST rate, domestic GST rate, invoice prefix, financial year start.

FR-D84 - Branding

The app shall allow admins to upload a logo, seal, and signature (PNG/JPEG, validated by magic bytes; logo ≤5MB, seal/signature ≤2MB), shown on printed invoices.

FR-D85 - User Management

The app shall allow admins to create staff users (within plan limit), change passwords, and deactivate users.

10.20 Printing

FR-D86 - Print Templates

The app shall provide invoice templates: Standard, Modern, Professional (A4), and Thermal (80mm), with Bhutanese theming.

FR-D87 - Print Sanitization

All printed HTML shall be sanitized (no scripts, iframes, event handlers, js-URIs) before loading in the print window.

FR-D88 - Report Printing

The app shall print reports (trial balance, P&L, balance sheet, etc.) via hidden BrowserWindow -> `webContents.print()`.

FR-D89 - Barcode Printing

The app shall print barcode label sheets.

11. Non-Functional Requirements

11.1 Performance

NFR-P01 - App Launch: The desktop app shall launch and show the main screen within 5 seconds on a mid-range Windows 10 PC.

NFR-P02 - Sale Completion: A sale (from clicking "Complete" to print dialog) shall complete within 2 seconds for a cart of ≤ 50 items.

NFR-P03 - Report Generation: Any report shall generate within 5 seconds for a database of up to 100,000 transactions.

NFR-P04 - Search: Item search at the POS shall return results within 300ms for an inventory of 10,000 items.

NFR-P05 - Server API: Server API endpoints shall respond within 2 seconds (p95) under normal load.

NFR-P06 - Backup: A local backup shall complete within 60 seconds for a 50 MB database.

NFR-P07 - Memory: The desktop app shall use less than 1 GB of RAM during normal operation.

11.2 Reliability

NFR-R01 - Offline Operation: The app shall operate fully offline for all daily operations (sales, inventory, reports, printing) for at least 7 days without internet.

NFR-R02 - No Data Loss: A sale or transaction shall not be committed until all related writes (transaction, lines, stock, GST, audit) succeed atomically.

NFR-R03 - Crash Recovery: On crash, the app shall recover the database from encrypted `.secure` or emergency backups without data loss.

NFR-R04 - Server Uptime: The server shall target 99.9% uptime (Vercel hosting).

NFR-R05 - DB Integrity: The audit chain and integrity seals shall be verified on every boot; any violation shall be logged and surfaced to the admin.

11.3 Usability

NFR-U01 - Onboarding Time: A non-technical user shall complete onboarding (install -> first sale) in ≤ 10 minutes.

NFR-U02 - Plain Language: All UI text shall be in simple English suitable for non-technical Bhutanese shop owners.

NFR-U03 - Error Messages: Error messages shall be human-readable and actionable (no stack traces shown to end-users).

NFR-U04 - Responsive: The desktop app UI shall be usable at 1200×700 resolution and above.

NFR-U05 - Notifications: Success/error feedback shall be shown via toast notifications (with audio chime, mutable).

11.4 Maintainability

NFR-M01 - Build Reproducibility: `npm run dist:win` shall produce a working installer from a clean `npm install` on a machine with the prerequisites.

NFR-M02 - Linting: The codebase shall pass `npm run lint` with no errors.

NFR-M03 - Modularity: Services shall be loosely coupled (each feature has its own service class) to allow independent updates.

NFR-M04 - Migration Scripts: Database schema changes shall be handled in code (auto-migrations on app start) or via documented Python scripts.

11.5 Scalability

NFR-S01 - Server: The server shall support up to 10,000 concurrent licenses and 100 RPS via Vercel + MongoDB Atlas.

NFR-S02 - Desktop: The desktop app shall support up to 100,000 transactions and 10,000 inventory items without significant degradation.

11.6 Localization

NFR-L01 - Currency: All amounts shall be displayed as Bhutanese Ngultrum (Nu. 1,234.56).

NFR-L02 - Dates: Dates shall be displayed in a readable format (e.g., 20 July 2026).

NFR-L03 - Address: Addresses shall use Bhutanese dzongkhag/gewog fields.

12. Security Requirements

See SOP Section 18 for complete security procedures and implementation details.

12.1 Server Security

SEC-S01 - Authentication: Admin login via NextAuth (JWT, 24h). POS login via custom JWT + refresh-token rotation with reuse detection.

SEC-S02 - Password Policy: Min 8 chars, upper + lower + digit + special. Bcrypt 12 rounds.

SEC-S03 - Account Lockout: 5 failed logins -> 15-min lockout.

SEC-S04 - Rate Limiting: Redis-backed with memory fallback. Limits: default 10/min, license 5/min, auth 5/15min, API 60/min.

SEC-S05 - Security Headers: CSP, HSTS (prod), X-Frame-Options DENY, X-Content-Type-Options nosniff, Referrer-Policy, Permissions-Policy.

SEC-S06 - Anti-Enumeration: Forgot-password and verification-status shall not reveal if an email exists.

SEC-S07 - License-Device Binding: HMAC-SHA256 binding using `LICENSE_BINDING_SECRET` (or `NEXTAUTH_SECRET`).

SEC-S08 - Audit Logging: 29 event types, 3 severity levels, stored in MongoDB.

SEC-S09 - Secrets: `NEXTAUTH_SECRET`, `LICENSE_BINDING_SECRET`, `ENCRYPTION_PEPPEER` shall be ≥ 32 chars and never committed to git.

SEC-S10 - Input Validation: All API inputs validated with Zod.

12.2 Desktop App Security

SEC-D01 - Electron Hardening: `contextIsolation: true, nodeIntegration: false, sandbox: true, webSecurity: true`.

SEC-D02 - Secure Preload: No raw Node APIs exposed. Only a sanitized `electronSecureAPI` via `contextBridge`. Every input sanitized (max length, control-char strip).

SEC-D03 - IPC Validation: Every IPC input validated with Zod. Every handler can require auth, admin, and rate limiting (5 req/15min per WebContents).

SEC-D04 - CSP: Strict CSP in production (`script-src 'self'`, no `unsafe-inline/unsafe-eval`).

SEC-D05 - URL Allowlists: `shell.openExternal` and SaaS requests limited to allowed domains (`jindapos.com`, `api.jinda.com`, `jinda.vercel.app`, `github.com`, Google Drive/MEGA).

SEC-D06 - SSRF Protection: SaaS requests shall block hosts that resolve to private IPs (DNS-rebinding guard).

SEC-D07 - DB Encryption: AES-256-CBC + HMAC at rest; plaintext securely deleted on close.

SEC-D08 - Integrity Seals: HMAC-SHA256 seals on GST rates, user roles, period locks. Verified on boot.

SEC-D09 - Audit Chain: Tamper-evident linked audit log, verified on boot.

SEC-D10 - Device ID Protection: Encrypted persistent fallback (`.device-id` AES-256-GCM) prevents cloning.

SEC-D11 - Image Upload Validation: Magic-byte validation (PNG `89 50`, JPEG `FF D8`) + size limits.

SEC-D12 - Print Sanitization: Print HTML stripped of scripts, iframes, event handlers, js-URIs.

SEC-D13 - Token Storage: Tokens in OS keychain (keytar) with encrypted-file fallback.

SEC-D14 - Path-Traversal Guards: File paths validated against traversal.

SEC-D15 - Window Hardening: `setWindowOpenHandler` denies new windows; `will-navigate` restricts to allowed origins; DevTools forced closed in production.

13. User Flows

See the **Standard Operating Procedure (SOP)** document for detailed **step-by-step procedures covering:** - New customer onboarding (SOP Section 11) - License activation and transfer (SOP Sections 11.3 - 11.7) - Daily sales operations (SOP Section 12) - GST filing (SOP Section 13) - Update publishing and distribution (SOP Sections 9 and 17)

>

The PRD defines *what* the system must support; the SOP defines *how* users perform these tasks.

14. Data Model

14.1 Server (MongoDB)

Collection	Key Fields	Purpose
admins	username (unique), passwordHash (bcrypt), role	Admin login
posusers	username, email (unique), password, isVerified, accountStatus, trialStart/End, licenseKey, deviceId, deviceFingerprint, failedLoginAttempts, lockedUntil, approvedDevices[], activeSessions[]	POS user accounts
licenses	licenseKey (unique, DTS-XXXX-XXXX-XXXX), customerName, email, companyName, plan, maxUsers, status, deviceId, activationSecret, expiryDate, activationCount, lastTransferAt, monthlyTransferCount, revoked* fields	Licenses
customers	name, email (unique), company, phone, address (street/gewog/dzongkhag)	Customer profiles
sessions	userId, refreshTokenHash (SHA-256), tokenFamily, deviceId, ipAddress, userAgent, isRevoked, expiresAt (TTL 7d)	Refresh-token sessions
tokens	userId/licenseId, token (SHA-256), type (verify/reset/license-otp/device-verification), attempts, expiresAt (TTL)	Verification/OTP tokens
trials	deviceId (unique), userId, startDate, duration (7d)	Trial tracking
updates	version (unique), notes, downloadUrl, isLatest, status, rolloutPercent, forced, fileId, fileSize, fileSha512, releaseDate	Software updates
auditlogs	eventType, severity, userId, email, ipAddress, userAgent, deviceId, sessionId, details, metadata	Security audit
contactmessages	name, email, subject, message, status, replyNote	Contact form

14.2 Desktop App (SQLite)

See SOP Section 7.3 for the complete list of desktop database tables.

>

The desktop app uses SQLite with **14 core tables** (users, companies, accounts, contacts, items, transactions, transaction_lines, stock_movements, invoices, invoice_items, gst_entries, audit_logs, settings, period_locks) and **20+ feature tables** (purchase_orders, quotations, refunds, expenses, recurring_transactions, held_carts, branches, employees, payroll_payments, barcodes, price_lists, integrity_hashes, and more).

14.3 Default Chart of Accounts

See SOP Section 7.5 for the complete default Chart of Accounts.

>

The system ships with a standard double-entry chart organized into 6 top-level groups: Assets (1000), Liabilities (2000), Equity (3000), Income (4000), COGS (5000), and Operating Expenses (6000), with 17 default sub-accounts covering cash, bank, debtors, creditors, GST input/output/payable, PF, TDS, capital, retained earnings, sales revenue, and common expenses.

15. API Specification

15.1 Server Endpoints (40 total)

Detailed in SOP Section 10.10. Summary:

- **Auth (15):** `/api/auth/{nextauth, register, login, logout, refresh, status, verify-email, resend-verification, verify-device, resend-device-code, verification-status, forgot-password, reset-password, change-password, captcha}`
- **License (3):** `/api/license/{verify, activate, revoke}`
- **Admin (8):** `/api/admin/{licenses, licenses/[id], licenses/extend, customers, customers/[id], users, messages, messages/[id]/reply, security/dashboard}`
- **Updates (4):** `/api/updates/{, [id], latest, upload} + /updates/latest.yml`
- **Other (6):** `/api/{download, health, contact, chat, terms, privacy, stats}`

15.2 Desktop App IPC (100+ channels)

Grouped: auth, dashboard, pos, inventory, contacts, accounts, transactions, gst, reports, backup, cloudBackup, settings, print, heldCarts, expenses, payroll, purchaseOrders, quotations, refunds, recurring, auditLogs, agedReports, barcodes, employees, branches, csvImport, splitPayment, tieredPricing, emailInvoice, license, update, posAuth, saas, secureStorage, shell, app.

All IPC handlers use a secure wrapper with optional auth/admin/validator/rate-limit enforcement and a uniform error envelope `{ ok, status, data: { success, error } }`.

16. Reporting and Analytics Requirements

16.1 Reports (Desktop)

Report	Type	Frequency
Trial Balance	Accounting	On demand
Profit & Loss	Accounting	On demand (period)
Balance Sheet	Accounting	On demand (as-of date)
Outstanding Receivables/Payables	Finance	On demand
Stock Report	Inventory	On demand
Sales Report	Sales	On demand (period)
Purchase Report	Procurement	On demand (period)

Report	Type	Frequency
Payroll Report	HR	On demand (period)
Customer Insights	Sales	On demand
Aged Receivables/Payables	Finance	On demand
GST Summary	Compliance	Monthly

16.2 Admin Dashboard Analytics (Server)

- License counts by status (active, inactive, expired, suspended, revoked).
- Plan breakdowns (Starter, Growth, Pro, Enterprise, Lifetime).
- Licenses expiring soon (next 30 days).
- Recent licenses created.
- Security alerts, login stats, active sessions, locked accounts, top failing IPs, suspicious users.

16.3 Export Formats

- **Excel (.xlsx)**: all reports and lists, Bhutanese-themed.
- **PDF (.pdf)**: all reports, Bhutanese-themed, with page numbers and footer.

17. Integration Requirements

Integration	Purpose	Tech
MongoDB Atlas	Server database	Mongoose 8
Redis	Rate limiting	ioredis (memory fallback)
Cloudflare R2 / AWS S3	Installer file storage (optional)	@aws-sdk/client-s3
GitHub Releases	Update file hosting	Public URLs
Gmail SMTP	Email sending	Nodemailer
OpenRouter	AI chatbot	Streaming SSE
Google Drive API	Cloud backup	googleapis (OAuth loopback)
MEGA API	Cloud backup	megajs
Vercel	Server hosting	Next.js 16
electron-updater	Auto-update	electron-updater 6
keytar	OS keychain token storage	keytar 7
better-sqlite3	Local database	better-sqlite3 12
xlsx (SheetJS)	Excel import/export	xlsx 0.18

Integration	Purpose	Tech
jsPDF	PDF export	jspdf 4 + jspdf-autotable 5

18. Platform and Compatibility

18.1 Desktop App

Platform	Support	Build Target
Windows 10/11 (x64)	Primary	NSIS + Portable
Windows 10/11 (ia32)	Secondary	NSIS
macOS	Supported	DMG + ZIP
Linux	Supported	ApplImage + DEB + RPM

18.2 Server

- Hosted on Vercel (Next.js 16).
- MongoDB Atlas (any region close to Bhutan).
- Redis (optional; Redis Cloud, Upstash, or self-hosted).
- Browsers: modern Chrome, Firefox, Safari, Edge for the public website and admin dashboard.

18.3 Minimum Hardware (Desktop)

- CPU: any modern x64 processor.
- RAM: 4 GB minimum, 8 GB recommended.
- Disk: 1 GB free (app) + space for database and backups.
- Printer: Windows-compatible A4 printer or 80mm thermal printer.

19. Branding and Localization

19.1 Branding

- **Product name:** Jinda.
- **AppId:** `com.jinda.app`.
- **Invoice prefix:** `DT` (configurable).
- **License key prefix:** `DTS`.
- **Color theme:** Bhutanese palette - crimson `#9B2335`, saffron `#E8A020`, gold, navy. Prayer-flag gradient bar on invoices and reports.
- **Fonts:** Cinzel (headings), Inter / Outfit (body) on prints.
- **Footer:** "Powered by PhoJaa95" on PDF reports.

19.2 Localization

- **Currency:** Bhutanese Ngultrum (`Nu. 1,234.56`).
- **GST:** 5% standard, 0% domestic (configurable in Settings).

- **Payment modes:** Cash, Bank, Credit, Card, UPI, mBOB, BNB, TPay, DrukPNB, BDBL, DKBank.
 - **Addresses:** street, gewog, dzongkhag (Bhutanese administrative divisions).
 - **Language:** English (Bhutan uses English in business). Dzongkha is not required in v1.0.
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20. Out of Scope

The following are **not** part of v1.0 and may be considered for future versions:

- 1 **Multi-machine sync:** Real-time data sync between multiple computers. (Jinda is single-machine offline-first.)
 - 2 **Cloud/web app:** A browser-based POS. (Jinda is a desktop app by design.)
 - 3 **Dzongkha language UI:** The UI is English-only in v1.0.
 - 4 **Online payments:** Processing card/UPI payments through the app. (The app records payment modes but does not process payments.)
 - 5 **E-commerce integration:** Shopify, WooCommerce, etc.
 - 6 **Multi-currency:** Only Ngultrum is supported.
 - 7 **Mobile app:** No iOS/Android app.
 - 8 **Multi-tenant cloud:** The server is a licensing platform, not a multi-tenant SaaS.
 - 9 **AI-driven insights:** Beyond the chatbot, no predictive analytics.
 - 10 **Restaurant mode:** Table management, kitchen display.
 - 11 **Loyalty programs:** Points, rewards.
 - 12 **Weighing scale integration:** Hardware scale integration.
 - 13 **Customer-facing display:** Second screen for the customer.
 - 14 **Real-time stock transfer between branches:** Branches are independent in v1.0.
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21. Future Enhancements

Candidate features for v2.0+ (not committed):

- 1 **Optional cloud sync** for multi-device businesses.
 - 2 **Dzongkha language UI** toggle.
 - 3 **Mobile companion app** for owners to view reports on the go.
 - 4 **Online payment integration** (mBOB, BNB Pay) for checkout.
 - 5 **Weighing scale integration** for grocery shops.
 - 6 **Customer loyalty program** (points, tiers).
 - 7 **Restaurant mode** (tables, KOT).
 - 8 **AI-driven insights** (sales forecasting, anomaly detection).
 - 9 **E-commerce integration** for online stores.
 - 10 **Multi-currency** support for border trade.
 - 11 **SMS notifications** (in addition to email).
 - 12 **Branch stock transfers**.
 - 13 **Third-party API** for accountants to pull data.
 - 14 **Backup to additional clouds** (Dropbox, OneDrive).
 - 15 **Tablet/POS hardware integration** (cash drawers, customer displays).
-

22. Acceptance Criteria

22.1 Installation & Onboarding

- ☐ A user can download **Jinda Setup.exe** from the website and install it on Windows 10/11.
- ☐ On first launch, the EULA appears and acceptance is saved.
- ☐ The user can start a 7-day trial without a license key.
- ☐ The user can create an initial admin account.
- ☐ Onboarding to first sale takes ≤ 10 minutes.

22.2 Licensing

- ☐ A license key **DTS-XXXX-XXXX-XXXX** can be created in the admin dashboard.
- ☐ Entering the key in the desktop app activates it and binds it to the device.
- ☐ A second activation on a different device triggers the transfer flow (password -> OTP).
- ☐ After 2 transfers in a month, a 3rd is rejected.
- ☐ An expired license shows the expired screen; an extended license reactivates.
- ☐ A revoked license is denied on the next check.

22.3 POS

- ☐ A cashier can add items to the cart by search and by barcode.
- ☐ GST is auto-calculated at 5% (or 0% for domestic).
- ☐ A sale can be paid with any of the 10 payment modes.
- ☐ Split payments work (two modes in one sale).
- ☐ A credit sale is blocked if the customer is over their credit limit.
- ☐ Held carts can be saved and resumed.
- ☐ An A4 invoice and a thermal receipt can be printed.
- ☐ Inventory decreases and the customer ledger updates after a sale.

22.4 Accounting & GST

- ☐ The trial balance balances (debits = credits).
- ☐ The P&L and Balance Sheet generate correctly.
- ☐ The GST summary for a month shows correct output, input, and net payable.
- ☐ Filing a GST return marks the period as filed.
- ☐ Voiding a transaction creates a reversal entry and excludes it from GST.
- ☐ Period close prevents edits to the locked period.

22.5 Reports & Exports

- ☐ All reports generate within 5 seconds for $\leq 100,000$ transactions.
- ☐ Excel export produces a valid **.xlsx** file with Bhutanese styling.
- ☐ PDF export produces a valid **.pdf** file with Bhutanese styling.
- ☐ CSV/Excel import works for contacts and items ($\leq 10,000$ rows).

22.6 Security

- ☐ The database is encrypted to **.secure** on close and decrypted on open.
- ☐ Integrity seals are verified on boot; tampering is logged.

- [] 5 failed logins lock the account for 15 minutes.
- [] Refresh-token reuse revokes the token family.
- [] The CSP header is present in production.
- [] No raw Node APIs are exposed to the renderer.

22.7 Backup & Updates

- [] A local backup can be created and restored (app relaunches).
- [] Auto-backup runs daily at 23:00.
- [] Google Drive and MEGA cloud backups work.
- [] An update published in the admin dashboard is detected within 4 hours.
- [] The SHA512 of the downloaded update is verified.
- [] Only admin users can install updates.

22.8 Server

- [] The admin can log in, create licenses, manage customers, and publish updates.
- [] The security dashboard shows alerts, login stats, and locked accounts.
- [] Rate limiting blocks excessive requests.
- [] Emails (verification, OTP, reset) are delivered.
- [] `/api/health` returns `{ status: "ok" }`.

Appendix A - Glossary

See SOP Appendix D for the complete shared glossary of technical terms (API, Electron, JWT, MongoDB, Next.js, SQLite, Vite, Zustand, etc.).

>

The following terms are specific to the product requirements context:

Term	Meaning
Activation Secret	A UUID returned only on first license activation, used for future handshakes
Activation Count	Number of times a license has been activated (incremented each verify)
Average Cost	Inventory cost method: weighted average of purchase prices
Chart of Accounts	The list of all accounting accounts (see SOP Section 7.5)
Double-Entry	Accounting method where every transaction has equal debits and credits
Integrity Seal	HMAC verifying a value hasn't been tampered with
License Binding	Tying a license to a device ID
Offline Grace	7-day period the app works without server verification
Period Close	Locking an accounting period

Term	Meaning
PRD	Product Requirements Document (this document)
Rollout %	Percentage of users who get an update
Seat Limit	Max users per plan
Trial Lock	Encrypted file preventing trial reset
Transaction Lines	Debit/credit lines in a double-entry transaction
Void	Reversing a transaction with a reason

End of Product Requirements Document

This PRD defines the complete requirements for the Jinda POS Platform v1.0 - covering the licensing/distribution server and the offline desktop POS & accounting app, built for Bhutanese businesses. All functional and non-functional requirements herein are testable and map to the acceptance criteria in Section 22.